Exercise Sheet 19

The notation is that of Lecture 19: $s_\lambda = a_{\lambda+\delta}/a_\delta$ in n variables, $c^{\nu}_{\lambda\mu}$ are the Littlewood–Richardson coefficients, and $L(a,b) = L(a\omega_1 + b\omega_2)$ for $\mathfrak{sl}_3$, with $3 = \mathbb{C}^3$. Exercises 19.1–19.3 develop the product rules. Exercises 19.4–19.6 treat determinant shifts, symmetric groups and invariants.

Exercise 19.1 (The dual Pieri rule). Let λ be a partition with $\ell(\lambda) \le n$, let $k \in \mathbb{Z}$ with $0 \le k \le n$, and let $e_k = \sum_{i_1<\cdots<i_k} x_{i_1}\cdots x_{i_k}$. For partitions $\lambda \subseteq \nu$ with at most n parts, call ν/λ a *vertical strip* if $\nu_i - \lambda_i \in \{0,1\}$ for all i.

(i) Prove that $s_\lambda e_k = \sum_\nu s_\nu$, summed over partitions ν with $\ell(\nu) \le n$ such that ν/λ is a vertical strip of size k.

(ii) Deduce the decomposition of $\mathbb{S}_\lambda(\mathbb{C}^n) \otimes \Lambda^k\mathbb{C}^n$.

(iii) Show that $\mathbb{S}_\lambda(\mathbb{C}^n) \otimes \Lambda^n\mathbb{C}^n \cong \mathbb{S}_{\lambda+(1^n)}(\mathbb{C}^n)$ as $\mathfrak{gl}_n$-modules. What does this say for $\mathfrak{sl}_n$?

Exercise 19.2 (Two products in $\mathfrak{sl}_3$).

(i) Use Pieri's rule to compute $s_{(1,1)}h_2$ for $n = 3$. Decompose $3^* \otimes 6$, where $3^* = \Lambda^2\mathbb{C}^3 = L(0,1)$ and $6 = \operatorname{Sym}^2\mathbb{C}^3 = L(2,0)$. Check dimensions.

(ii) Use the Littlewood–Richardson rule, listing every tableau, to compute $s_{(2,1)}s_{(1,1)}$ for $n \ge 4$. Deduce the decomposition of $8 \otimes 3^*$ for $\mathfrak{sl}_3$. Compare with Exercise 19.1.

Exercise 19.3 (One coefficient computed two ways). Compute $c^{(4,2,1)}_{(3,1),(2,1)}$ by filling $(4,2,1)/(3,1)$ with content $(2,1)$. Then compute $c^{(4,2,1)}_{(2,1),(3,1)}$ by filling $(4,2,1)/(2,1)$ with content $(3,1)$. Interpret the answer for $\mathfrak{sl}_3$.

Exercise 19.4 (Determinant shifts after tensoring).

(i) Let $n \ge 3$. For $\mathfrak{sl}_n$, $(\mathbb{C}^n)^* \cong L(\omega_{n-1})$ is represented by (1^{n-1}). Use Pieri's rule to decompose $\mathbb{C}^n \otimes (\mathbb{C}^n)^*$ for $\mathfrak{sl}_n$. Explain why the corresponding $\mathfrak{gl}_n$ calculation describes $\det \otimes \mathbb{C}^n \otimes (\mathbb{C}^n)^*$.

(ii) Decompose $\Lambda^2\mathbb{C}^3 \otimes \Lambda^2\mathbb{C}^3$ for $\mathfrak{gl}_3$. Deduce the decomposition of $3^* \otimes 3^*$ for $\mathfrak{sl}_3$, and identify its symmetric and alternating parts.

(iii) Decompose $(\mathbb{C}^3)^{\otimes 3}$ for $\mathfrak{sl}_3$. Find a nonzero $\mathfrak{sl}_3$-fixed tensor and say why it is not $\mathfrak{gl}_3$-fixed.

Exercise 19.5 (Outer and inner products).

(i) Decompose the inner tensor product $V^{(2,1)} \otimes V^{(2,1)}$ of S_3-modules using characters.

(ii) Using the outer-product remark of Lecture 19 and Step 1 of its $\mathfrak{sl}_3$ calculation, decompose $\operatorname{Ind}_{S_3\times S_3}^{S_6}(V^{(2,1)} \boxtimes V^{(2,1)})$. Check the dimension with the hook-length formula.

(iii) Which of these two products controls $\mathbb{S}_{(2,1)}(\mathbb{C}^n) \otimes \mathbb{S}_{(2,1)}(\mathbb{C}^n)$?

Exercise 19.6 (An invariant cubic on $\mathfrak{sl}_3$). An *invariant trilinear form* on $\mathfrak{g} = \mathfrak{sl}_3$ is a trilinear map $F\colon \mathfrak{g}^3 \to \mathbb{C}$ with

$$F([W,X],Y,Z) + F(X,[W,Y],Z) + F(X,Y,[W,Z]) = 0$$

for all $W, X, Y, Z \in \mathfrak{g}$. Equivalently, $F \in \operatorname{Hom}_{\mathfrak{g}}(8 \otimes 8 \otimes 8, \mathbb{C})$.

(i) Prove that the space of invariant trilinear forms has dimension 2.

(ii) Show that $f(X,Y,Z) = \operatorname{tr}(X[Y,Z])$ and $d(X,Y,Z) = \operatorname{tr}(X(YZ+ZY))$ form a basis, with f alternating and d symmetric.

(iii) Deduce that every invariant cubic $p(X) = F(X,X,X)$, with F symmetric and invariant, is a multiple of $\det X$.